

Understanding In-context Learning of Addition via Activation Subspaces

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Abstract

To perform few-shot learning, language models extract signals from a few input-label pairs, aggregate these into a learned prediction rule, and apply this rule to new inputs. How is this implemented in the forward pass of modern transformer models? To explore this question, we study a structured family of few-shot learning tasks for which the true prediction rule is to add an integer k to the input. We introduce a novel method that localizes the model’s few-shot ability to only a few attention heads. We find that this method and the localization finding generalize to four additional task families spanning arithmetic and semantic tasks. We then perform an in-depth analysis of individual heads, via dimensionality reduction and decomposition. As an example, on Llama-3-8B-instruct, we reduce its mechanism on our tasks to just three attention heads with six-dimensional subspaces, where four dimensions track the unit digit with trigonometric functions at periods 2, 5, and 10, and two dimensions track magnitude with low-frequency components. To deepen our understanding of the mechanism, we also derive a mathematical identity relating “aggregation” and “extraction” subspaces for attention heads, allowing us to track the flow of information from individual examples to a final aggregated concept. Our results demonstrate how tracking low-dimensional subspaces of localized heads across a forward pass can provide insight into fine-grained computational structures in language models. Our code is available at <https://github.com/xyVickyHu/addition-subspaces>.

1 Introduction

Large language models (LLMs) exhibit in-context learning (ICL) abilities; for instance, they can few-shot learn new tasks from a small number of demonstrations in the prompt. To understand this ability, past works have constructed detailed models of ICL for small synthetic language models (Garg et al., 2022; Akyürek et al., 2023; von Oswald et al., 2023) as well as coarser-grained analyses of large pretrained models (Olsson et al., 2022; Hendel et al., 2023; Todd et al., 2024). However, little is known about the fine-grained computational structure of ICL for large models.

ICL extracts task information from demonstrations and applies the aggregated information to input queries. Previous work (Todd et al., 2024) constructed a vector (i.e., the “function” vector) from demonstrations that encodes the task information—for instance, adding it to the residual stream on zero-shot inputs recovers ICL behavior. However, two questions remain elusive: (1) How precisely do function vectors encode task information? (2) How do models aggregate information from ICL examples to form such function vectors?

To address these questions, we first develop a novel general pipeline for localizing few-shot ability to a small number of attention heads and low-dimensional subspaces, and furthermore perform a case study for arithmetic tasks (i.e., learning to add a constant k to the input) for mechanistic depth. This family of tasks has the advantage of providing a large number of tasks (different integers k) that all share the same input domain, which is important

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to rule out domain-based shortcuts when analyzing ICL mechanisms (§2.1). Arithmetic also has the advantage of being well-studied in other (non-in-context) settings (Zhou et al., 2024; Kantamneni & Tegmark, 2025), allowing us to situate our results with other known mechanisms. Finally, most LLMs perform this task reliably (e.g. 87% accuracy for Llama-3 and 90% for Qwen-2.5).

We first introduce a head-localization pipeline that combines a novel sparse joint optimization objective with statistical and effect-size refinements. Our approach is inspired by function vectors (Todd et al., 2024), which mimic ICL behavior when patched into the residual stream (§2.2). Whereas Todd et al. (2024) rank heads independently by their *individual* effects on ICL performance, we jointly optimize coefficients $c_h \in [0, 1]$ over all heads to identify a sparse, weighted *combination* of head outputs that maximizes intervention accuracy (§3.1). Our method successfully finds a small number of heads that recover most of the ICL performance (§3.2); for example, for addition tasks, Llama-3-8B-instruct achieves 68% accuracy using function vectors from just three heads (78% of the original ICL accuracy). Applying the same pipeline to subtraction, multiplication, and the abstractive and extractive tasks from Todd et al. (2024) reveals shared structure alongside task-dependent localization: one head in a middle layer is selected as a main head across all five families, several additional heads recur across arithmetic tasks, and the semantic families rely on broader head sets (Appendix B).

We next perform a detailed analysis of these heads, through dimensionality reduction and decomposition. We find that the task information in the selected heads is encoded in low-dimensional subspaces across task families. For addition, which we study in depth, these subspaces exhibit an interpretable trigonometric structure. For each of the three main heads in Llama-3-8B-instruct, we get a 6-dimensional subspace, where 4 dimensions encode periodic functions with periods 2, 5, and 10 (tracking the units digit) and 2 dimensions encode low-frequency functions (tracking the tens digit) (§4). This interestingly aligns with recent findings in non-in-context settings where addition is also encoded by trigonometric (Zhou et al., 2024) or helical functions (Kantamneni & Tegmark, 2025), suggesting a deeper relation between non-ICL and ICL machinery.

Finally, we further study the flow of information across tokens, by deriving a general mathematical relation between “extractor” and “aggregator” subspaces, building on the second-order logit lens (Gandelsman et al., 2025). This lets us study the task-related information that heads extract from each token. For example, in Llama-3-8B-instruct we find that task information is mostly extracted from label tokens (§5.2).

In summary, we found that task-specific information in ICL can be localized to a small number of attention heads and low-dimensional subspaces with structured patterns. We also developed a way to trace the task information flowing from the “extractor” subspaces to the “aggregator” subspaces. Our findings show how even in large models, ICL mechanisms can be localized to specialized activation subspaces from a small number of heads that extract, represent, and aggregate information in interpretable ways. Our key contributions are: (1) a novel method to identify relevant attention heads for ICL; (2) a precise mathematical relation between signals from earlier tokens and the output token; and (3) a focused analysis of ICL addition that reverse-engineers rich latent structures and sophisticated computational strategies in LLMs.

2 Preliminaries

2.1 Model and Task

Model. We focus on Llama-3-8B-instruct (Grattafiori et al., 2024) in the main body, with 32 layers, 32 attention heads per layer, and residual dimension 4096. We denote each head as a tuple (layer index, head index); both indices range from 0 to 31. To broaden our analysis in model size, training type, and model family, we also experiment on Phi-4-14B (Abdin et al., 2024) and Qwen-2.5-7B (Yang et al., 2024) and report results in the appendix. The results are consistent for the Phi model, and directionally similar for the Qwen model.

Task. We consider five families of ICL tasks: three arithmetic families (addition, subtraction, and multiplication) and the abstractive and extractive families of [Todd et al. \(2024\)](#), which include abstractive tasks like product-company (“iPhone 5→Apple”) and extractive tasks like animal-vs-object (“bear, ukulele, blanket→bear”); Table 2 in Appendix B.1 specifies all five families. Our localization pipeline applies to each of them (Appendix B); for mechanistic depth, we center our case study on addition, a structured family of ICL tasks that we call *add-k*. For a constant k , the *add-k* task is to add k to a given input integer x to predict $y = x + k$. In an n -shot ICL prompt, the model is given n demonstrations of the form “ $x_i \rightarrow y_i$ ” with $y_i = x_i + k$, concatenated using the separator “#”, followed by a query “ $x_q \rightarrow$ ” (see Figure 3 in Appendix for an illustration). A key advantage of this family of tasks is that ICL prompts for different tasks only differ in k (i.e. $y_i - x_i$) but not in the input domain, enabling us to isolate task information from input content so as to dissect the ICL mechanism at a finer granularity.

This advantage motivates our focus on *add-k*: in semantic tasks such as product-company, the input domain itself leaks information about the task ([Min et al., 2022](#)), making it difficult to distinguish whether success stems from extracting the task rule or from leveraging domain-specific associations. For *add-k*, the query x_q alone provides no information about k , so the model must infer k from the demonstrations, which cleanly isolates the key components of the ICL mechanism. Moreover, addition is well studied in non-in-context settings ([Zhou et al., 2024](#); [Kantamneni & Tegmark, 2025](#)), so studying it under ICL fills a gap and allows us to compare our findings with the known non-ICL mechanisms.

We construct data for the *add-k* task by varying $x \in [1, 100]$ and $k \in [1, 30]$ (thus $y \in [2, 130]$) since Llama-3 models can reliably solve addition in this range. We consider three types of prompts: (1) *five-shot ICL prompt*, where all five demonstrations satisfy $y_i = x_i + k$ for a fixed $k \in [1, 30]$ (also called *add-k ICL*); (2) *mixed-k ICL prompt*, where the demonstrations are $y_i = x_i + k_i$ for possibly different k_i values; and (3) *zero-shot prompt*, where there are no demonstrations: the prompt is “ $x_q \rightarrow$ ” for some x_q . We mostly study five-shot ICL prompts and use them to generate function vectors (§2.2). We also examine signals extracted from demonstrations in mixed- k ICL prompts, a variant with mistaken demonstrations, in §5.2. We use zero-shot prompts to evaluate heads and function vectors (§2.2).

2.2 Activation Patching and Function Vectors

Next, we briefly review *activation patching*, a common interpretability technique that will be used throughout the paper, and the *function vector*, a construction that we will use to identify important heads for our ICL tasks.

Activation patching. Activation patching is performed by taking the activations of a model component when the model is run on one prompt, then patching in these activations when the model is run on a different prompt. Patching can either *replace* the model’s base activations or *add* to them; we will primarily consider the latter.

Specifically, if z_l is the original value of the residual stream at layer l at the final token position “→”, we patch in the replacement $z_l + v$, where v is constructed from activations on a different input prompt. We choose the layer l at one-third of the network’s depth and construct v from “function vector” heads, following [Todd et al. \(2024\)](#), as described next.

Function vectors. Function vectors v_k are vectors constructed from the outputs of selected attention heads, designed so that adding v_k to the residual stream of a zero-shot prompt approximates the effect of the five-shot *add-k* task. For example, v_k might be the average output of one or more attention heads across a set of five-shot *add-k* examples. We define the *intervention accuracy* of v_k as the average accuracy obtained on zero-shot prompts when v_k is added to the residual stream across all tasks k . Because k is independent of the input queries, this metric captures how effectively v_k encodes task-specific information about k . For comparison, we define *clean accuracy* as the accuracy on five-shot prompts without any intervention, also averaged across all tasks k .

In more detail, consider an attention head h : let $h(p)$ denote the output of head h on prompt p at the last token position, and define h_k as the average of $h(p)$ across all five-shot

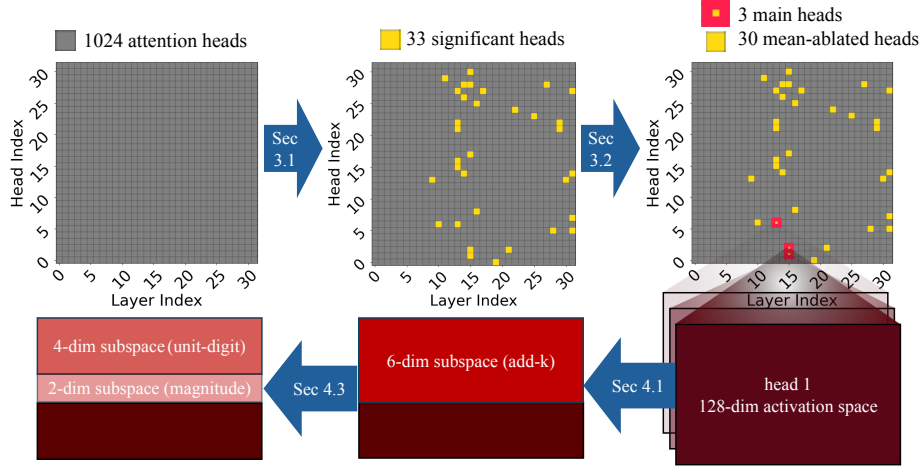


Figure 1: The chain of localization in §3 and §4. We first identify 33 significant attention heads (out of 1024) via a global optimization method (§3.1), then narrow down to 3 main heads while mean-ablating the remaining 30 (§3.2). We next study the structure of the representation of each main head by localizing it to a six-dimensional subspace (§4.1), and decompose it into a four-dimensional subspace encoding the unit digit and a two-dimensional subspace encoding the coarse magnitude (i.e., the tens digit) (§4.3).

add-k prompts.¹ Todd et al. (2024) identified a subset $\mathcal{H}$ of attention heads (for a different set of tasks) such that the vector $v_k = \sum_{h \in \mathcal{H}} h_k$ has high intervention accuracy—that is, adding it to the residual stream of zero-shot prompts effectively recovers few-shot performance.

Building on this framework, we consider multiple ways to construct v_k : (1) the task-specific mean h_k over the *add-k* task (as described above); (2) the overall mean $\bar{h}$ (i.e., average across all k), which removes task-specific information about k ; or (3) the specific value $h(p)$ on an individual prompt p . Throughout the paper, we call $\bar{h}$ the *mean-ablation* of head h and call h_k the *head vector* of h with respect to k . Beyond these three choices, we sometimes project a head’s output onto a lower-dimensional subspace or scale it by a coefficient. Unlike Todd et al. (2024), we recover a different set of heads by solving a novel global optimization problem (§3.1) and systematic ablation studies (§3.2), achieving higher performance.

3 Identifying Three Aggregator Heads

To understand the mechanism behind the *add-k* ICL task, we first need to find out what model components are responsible for performing it. In this section, we find that three heads do most of the work for *add-k*, by first solving an optimization problem to identify 33 significant heads out of 1024 (§3.1), and then narrowing down to three main heads via systematic ablations (§3.2). This process is illustrated in the first row of Figure 1. §3.3 then presents the two head-set refinement criteria behind this narrowing in their general form, which we reuse across task families, models, and prompt formats (Appendix B).

3.1 Identifying Significant Heads via Sparse Optimization

We will search for a set of heads that store the information for the *add-k* task, in the sense that their output activations yield good function vectors for *add-k* (§2.2).

¹Similar to Todd et al. (2024), we approximate this average using 100 randomly generated five-shot *add-k* prompts, where each prompt contains random demonstration inputs x_i and the final query input x_q is chosen exactly once from each integer in $[1, 100]$.

Formulating the sparse optimization. More formally, define $v_k(c) = \sum_h c_h \cdot h_k$, the sum of head outputs weighted by c , where h goes over all 1024 heads in the model. (Recall that h_k is the average output of head h averaged across a large dataset of five-shot *add-k* prompts.) We will search for a sparse coefficient vector $c \in [0, 1]^{1024}$ such that adding $v_k(c)$ to the residual stream of a zero-shot prompt achieves high accuracy on the *add-k* task.

Let $\ell(x_q, y_q; v)$ be the cross-entropy loss between y_q and the model output when intervening v onto the input “ $x_q \rightarrow$ ” (i.e. replacing z_l with $z_l + v$ in the forward pass on “ $x_q \rightarrow$ ”, where z_l is the layer- l residual stream at the last token). We optimize c with respect to the loss

$$\mathcal{L}(c) = \mathbb{E}_{k, x_q} [\ell(x_q, x_q + k; v_k(c))] + \lambda \|c\|_1, \quad (1)$$

where the regularization term with weight λ promotes sparsity.

Training details. We randomly select 25 *add-k* tasks for training and in-distribution testing, using the remaining five for out-of-distribution testing. For each task, we generate 100 zero-shot prompts “ $x_q \rightarrow$ ” with $x_q \in [1, 100]$ and target $x_q + k$. We split the 25 tasks’ data into train/val/test sets in proportions 0.7/0.15/0.15. We use AdamW with learning rate 0.01, batch size 128, and regularization $\lambda = 0.05$, clipping c to $[0, 1]$ after each gradient step.

Results: selecting significant heads. We get coefficients that achieve high intervention accuracy. In particular, on the five held-out tasks, the function vector weighted by the final-epoch coefficients achieves an intervention accuracy of 0.812, close to the clean accuracy of 0.870. To identify important heads, we inspect the coefficient values at the end of training for each layer and head index. We find 33 heads have coefficients c_h greater than 0.2, most (21) of which are one. In contrast, the other (991) coefficients are all smaller than 0.03, most (889) of which are zero. We call these 33 heads *significant heads*, denoted $\mathcal{H}_{\text{sig}}$.

More generally, we do not hand-pick this subset: we sort the trained coefficients (averaged over the last three checkpoints to smooth end-of-training jitter) in decreasing order and cut at the largest gap between consecutive values, keeping the heads above it. This parameter-free rule recovers exactly these 33 heads for *add-k*, and is the selection rule we apply to every other task family, model, and prompt format we study (the shot-count variants reuse the five-shot selection); those results appear in Appendix B.

Comparison to previous approach. We compare our optimization approach with the previous method from Todd et al. (2024) for identifying important heads, which selects heads based on average indirect effect (AIE). To perform a fair comparison, we construct our function vector by summing the outputs of our selected heads directly without weighting by coefficients, matching their methodology. Using this construction, we achieve an intervention accuracy of 0.798, close to the clean accuracy of 0.870, indicating that our 33 heads capture most of the necessary information for the *add-k* task. In contrast, selecting the top 33 heads according to AIE yields a much lower intervention accuracy of 0.414.²

3.2 Further Narrowing to Three Heads via Ablations

Mean-ablation as a task-agnostic base. We suspect many heads primarily store task-family-level signals (such as ensuring the output appears as a number) rather than encoding information about k itself. To test this, we perform *mean-ablations*: replacing each task-specific head vector h_k with the cross-task mean $\bar{h}$ across tasks in the family (§2.2).

Specifically, we conduct mean-ablations over subsets of the significant heads and measure the resulting intervention accuracy. Formally, for a retained set $\mathcal{S} \subseteq \mathcal{H}_{\text{sig}}$, we mean-ablate its complement $\mathcal{H}_0 = \mathcal{H}_{\text{sig}} \setminus \mathcal{S}$ and construct

$$v_k(\mathcal{S}) = \sum_{h \in \mathcal{S}} h_k + \sum_{h \in \mathcal{H}_{\text{sig}} \setminus \mathcal{S}} \bar{h}. \quad (2)$$

Thus, only heads in $\mathcal{S}$ retain variation specific to k , while all other significant heads contribute a task-agnostic base that is fixed across tasks. We use this construction both to compare different head sets and to isolate each head’s additional task-specific contribution.

²Todd et al. select roughly the top 2% of heads in their work ($k = 20$ for Llama-3-8B-instruct), which yields an even lower accuracy of 0.124 in our setting.

Recovery scan for each significant head. To isolate the incremental contribution of each head in $\mathcal{H}_{\text{sig}}$, we scale its coefficient up while mean-ablating other significant heads. For head h and integer scale $c \in \{0, \dots, 20\}$, we intervene with

$$v_k^{(h,c)} = c h_k + \sum_{g \in \mathcal{H}_{\text{sig}} \setminus \{h\}} \bar{g}. \quad (3)$$

Let $A_h(c)$ denote the resulting intervention accuracy on zero-shot prompts. At $c = 0$, the candidate head h is omitted while every other significant head contributes its cross-task mean, yielding a leave-one-out baseline. Increasing $c > 0$ then restores and scales only head h 's task-conditioned signal.

Crucially, this scan is performed on top of the family-shared means of the jointly selected significant heads, whereas [Todd et al. \(2024\)](#) rank heads independently over all attention heads. Our candidate starts from a behaviorally meaningful base, making its task-specific effect stand out more easily and reliably.

On add- k , three heads are sharply separated from every other significant head under this scan. Scaled heads (15,2) and (15,1) nearly match the clean accuracy of 0.87, reaching 0.866 and 0.817, respectively, while (13,6) reaches 0.693; the best other significant head reaches at most 0.170 at its optimal scale. We call these three heads *main* heads and zoom into their output spaces in later sections.

Validating necessity of main heads in the five-shot setting. We next test the necessity of the three main heads in the original five-shot ICL setting: mean-ablating their outputs on randomly generated five-shot prompts yields an accuracy of 0.43, half the clean accuracy of 0.87. For comparison, mean-ablating 20 random sets of three other significant heads yields accuracies close to clean: 95% have accuracy at least 0.86.

3.3 Two general head-set refinement criteria

The recovery scan of §3.2 supports two refinement criteria of independent interest, which we state here in general form and reuse across task families and models (Appendix B).

Recovery-positive heads: statistically supporting recovery. For each positive coefficient c , we compare per-example correctness against $c = 0$ on the same prompts. Let b be the number of examples fixed by the intervention and d the number broken. We use a one-sided exact McNemar test—equivalently, an exact binomial test on the $b + d$ discordant examples ([McNemar, 1947](#)). We apply a Bonferroni correction over the positive coefficients within each head, then control the false discovery rate across heads using the Benjamini–Hochberg procedure at $q = 0.05$ ([Benjamini & Hochberg, 1995](#)). We call a head *recovery-positive* if at least one positive coefficient has $b > d$ and survives both statistical corrections.

The paired comparison accounts for variation in prompt difficulty, while the two corrections protect against selecting a head merely because we scan many coefficients and many candidates. On Llama-3 add- k , this procedure narrows 33 significant heads to 13 recovery-positive heads, while intervention accuracy decreases only from 0.798 to 0.774. The compression is more dramatic for Qwen2.5-7B add- k : 230 significant heads narrow to 18, with accuracy 0.358 versus 0.368 for the full significant set (full results in Appendix B.2).

Main heads: materially dominant recovery. The paired tests above are designed to detect statistically reliable effects using per-example outcomes and may therefore retain heads with small overall gains. For a compact mechanistic core, we instead compare each head's aggregated recovery gain

$$G_h = \max_{c \in \{0, \dots, 20\}} A_h(c) - A_h(0).$$

Formally, let $G^* = \max_{h \in \mathcal{H}_{\text{sig}}} G_h$. For any task family with clean accuracy at least 0.25, we call a head *main* if $G_h \geq G^*/4$. On add- k , this rule recovers exactly the three heads of §3.2.

Beyond addition, we find one head (15, 1) in Llama-3-8B is a main head for every task family we study. Addition, subtraction, and multiplication are localized to 3, 4, and 5 main heads,

Head set	Selection criterion	# Heads	Accuracy [95% CI]
Significant	Selected by joint sparse optimization	33	0.798 [0.760, 0.832]
Recovery-positive	Reliably fixes more examples than it breaks	13	0.774 [0.736, 0.810]
Main	Recovery gain of at least 1/4 of the best-head gain	3	0.680 [0.640, 0.720]
AIE top-33	Ranked by average indirect effect	33	0.414 [0.372, 0.456]

Table 1: Nested head sets for Llama-3-8B-Instruct $\text{add-}k$, plus the AIE baseline of Todd et al. (2024) at the matched budget. Significant-set accuracy uses all 33 task-conditioned heads; the recovery and main rows keep the listed set task-conditioned and mean-ablate remaining significant heads. Brackets are 95% bootstrap confidence intervals. For reference, the clean five-shot ICL accuracy is 0.870 [0.838, 0.898] and the corrupted accuracy (zero-shot, every significant head mean-ablated) is 0.037 [0.030, 0.044].

respectively, with substantial reuse of the three main heads for addition. Phi-4 mirrors this structure: one head, (18, 12), appears in every family’s main set, and its subtraction main heads are exactly its addition main heads plus one extra head — the analogue of subtraction main heads = addition trio + (15, 17) in Llama-3 (Appendices B.1, B.2). The selection is also robust along the prompt-format and shot-count axes: the three $\text{add-}k$ main heads are selected under every prompt format and at every $n \in \{1, \dots, 10\}$ (Appendix B.3, B.4). Moreover, in every task family, model, prompt format, and shot count we consider, the main heads achieve higher intervention accuracy than the top- k head sets selected by Todd et al. (2024)’s AIE ranking while using far fewer heads (Appendix B.5).

Comparison of the two refinements. The two criteria serve different purposes: recovery-positive selection retains all statistically supported contributors and aims to preserve as much of the significant set’s recovery as possible, whereas main-head selection isolates a small number of materially dominant heads for subsequent mechanistic analysis. Table 1 summarizes the resulting nested sets and compares them with the head-selection method of prior work (Todd et al., 2024).

4 Characterizing the Aggregator Subspace

For the model to perform $\text{add-}k$, it must infer the task information (the number k) from the ICL demonstrations. Our next goal is to understand how task information is represented in the activation space. Since the three aggregator heads we identified carry most of this information, we can focus on analyzing their representation space.

In this section, we dissect their activation spaces in three stages (illustrated in Figure 2): (1) **Localize** a six-dimensional task subspace in each head via principal component analysis (PCA) (§4.1); (2) **Rotate** this subspace into orthogonal *feature directions* aligned with sinusoidal patterns across k (§4.2); (3) **Decompose** the six-dimensional space into a four-dimensional *unit-digit* subspace and a two-dimensional *magnitude* subspace that separately encode the units and tens of the answer (§4.3).

4.1 Localizing to Six-dimensional Subspace

To reduce the 128-dimensional activation space per head to a more tractable space, we first perform PCA on the 30 task vectors and find that just six directions explain 97% of the task variance (Figure 2a). We then check that the function vectors found earlier remain effective after projecting onto the subspace. Specifically, we replace each head vector h_k with its projection onto the subspace $\tilde{h}_k$ to obtain a new function vector $\tilde{v}_k = \sum_{h \in \mathcal{H}} \tilde{h}_k$ (a variant of $v_k = \sum_{h \in \mathcal{H}} h_k$ in §2.2). We find that $\tilde{v}_k$ has intervention accuracy 0.636, which is close to the intervention accuracy of 0.680 before projection (Table 1). Thus, we confine our study to the concise six-dimensional subspace for each head. Both the low dimensionality and the causal sufficiency of the projection generalize across the task families, models, prompt formats, and shot counts we study (Appendix B.6).

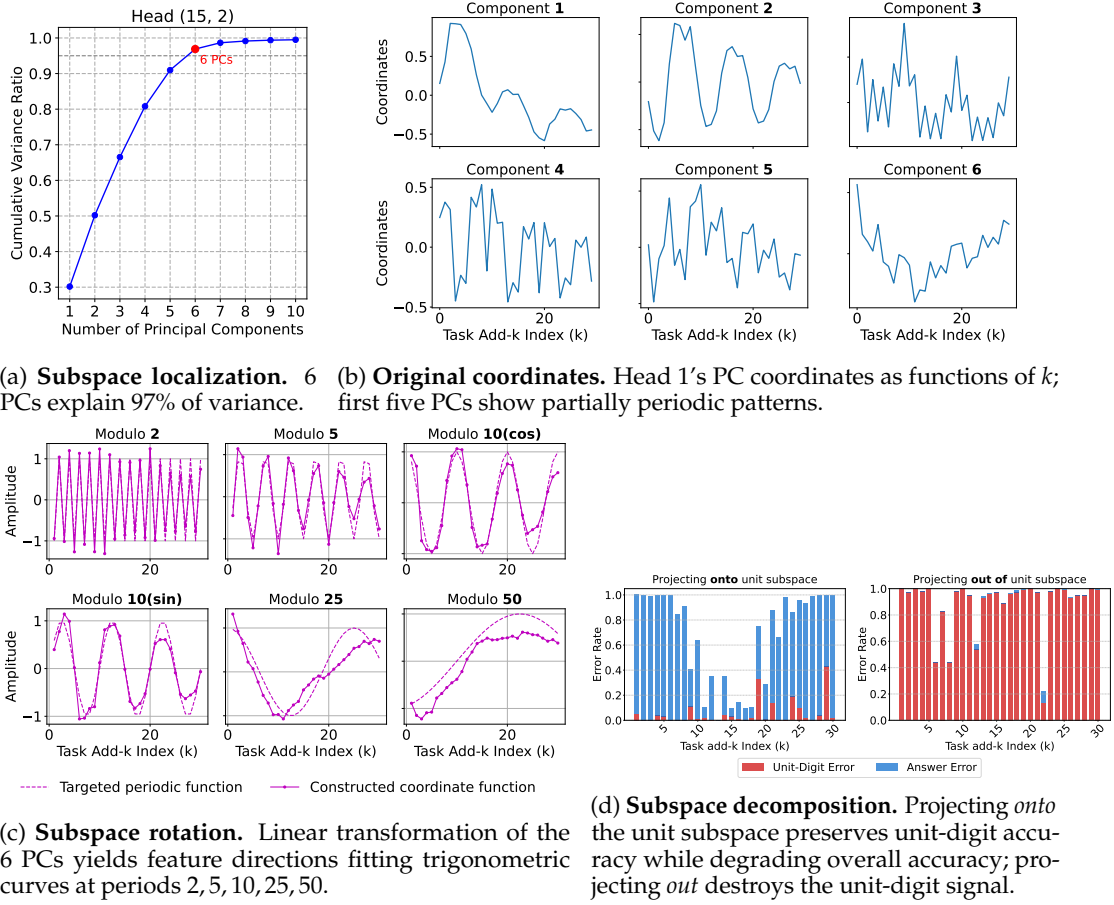


Figure 2: Characterizing the aggregator subspace, illustrated on head 1, (15, 2) (other heads and models in Appendices C–E): (a) PCA localizes task information to a 6-d subspace; (b) PC coordinates show periodic dependence on k ; (c) rotated feature directions encode clean trigonometric patterns; (d) causal steering validates the unit-digit subspace decomposition.

4.2 Identifying Feature Directions Encoding Periodic Patterns

To understand how each head’s six-dimensional subspace represents task information, we examine head vector coordinates (inner products with principal components (PCs)) as functions of k , revealing partially periodic patterns in the first five PCs (Figure 2b).

This motivates us to linearly transform the six PCs to find directions encoding pure periodic patterns. Mathematically, if we find a linear transformation of the six PC-coordinate functions that fits trigonometric functions, then applying the transformation to the PCs yields six directions whose coordinate functions encode the periodicity.

To find trigonometric functions to fit, we searched over different periods and phases and performed least squares regression. We found six trigonometric functions at periods 2, 5, 10, 10, 25, and 50 expressible as functions of the top 6 PCs with low regression error (Figure 2c). We apply the resulting linear transformation to the six PCs to obtain directions encoding these six pure periodic patterns, which we call *feature directions*.

4.3 Decomposing to Subspaces Encoding Subsignals

Leveraging the feature directions identified previously, we decompose the head activation subspace into lower-dimensional components that separately encode different subsignals relevant to the task—in this case, the units digit and tens digit.

By construction, the coordinate function of each feature direction (viewed as linear projections of h_k) is a periodic function of k . Mathematically, a feature direction with period T carries task information from the head vectors with “modulo T ”. Based on this, we hypothesize: (i) the feature direction corresponding to period two, which we call the “parity direction”, encodes the parity of k in *add-k* task; (ii) the subspace spanned by the feature directions with periods 2, 5, 10, which we call the “unit subspace”, encodes the unit digit of k ; (iii) the subspace spanned by the directions with periods 25, 50, which we call the “magnitude subspace”, encodes the coarse magnitude (i.e., the tens digit) of k .

We verify these hypotheses through causal intervention. We establish (1) **sufficiency** by showing that projecting a head vector *onto* the subspace preserves the relevant task signal; and (2) **necessity** by showing that projecting a head vector *out of* the subspace (i.e., onto its orthogonal complement) destroys the relevant task signal. We show experimental results for the unit-digit subspace in Figure 2d and defer complete results to Appendix E.

5 Signal Extractors of ICL Demonstrations

Having localized most of the task signal to three heads and their six-dimensional subspaces, we now analyze *how* the model extracts it from the ICL demonstrations. We find: (1) the label tokens are the main transport channel, carrying 96–99% of each main head’s task signal (§5.2); and (2) each demonstration $x_i \rightarrow y_i$ contributes a signal $y_i - x_i$ in the subspace.

5.1 Tracing the Subspace back to Previous Tokens

We begin with a mathematical observation that lets us trace the subspace at the final token back to corresponding subspaces at earlier tokens. A head h ’s output at the last token of a prompt p decomposes linearly over the prompt’s tokens: $h(p) = \sum_{t \in p} \alpha_t O_h V_h z_t$, where α_t is the attention weight from the last token to each token t , satisfying $\sum_{t \in p} \alpha_t = 1$, z_t is the residual stream input to the head h at token t , and V_h and O_h are the head’s value and output matrices. Intuitively, the head’s output is an attention-weighted sum of transformed residual streams from the previous tokens, so the signal extracted from each token t is its transformed residual stream $O_h V_h z_t$.

Letting W_h denote the projection onto the head’s six-dimensional subspace (§4.1), the projected task signal at the final token, $W_h h(p)$, thus decomposes into per-token contributions $\alpha_t W_h O_h V_h z_t$, each lying in the image of $W_h O_h V_h$.

5.2 Signal extraction from each demonstration

To understand how demonstrations contribute to the model’s generation at the last token, we identify which tokens contribute most, then examine what information they provide. By the analysis above, each token’s task-signal contribution through head h is $\alpha_t W_h O_h V_h z_t$, which factors into (1) **extracted information**: $W_h O_h V_h z_t$, the residual stream input projected into the relevant subspace; and (2) **aggregation weight**: the attention weight α_t . On a random mixed- k ICL prompt (with possibly conflicting k_i values), both the norm of the extracted information and the aggregation weight peak at the label tokens y_i (Appendix F.1).

To quantify this on standard five-shot prompts, we project each token’s contribution $\alpha_t O_h V_h z_t$ and the total output $h(p)$ onto the head vector h_k of the prompt’s task; the ratios are signed per-token shares that sum to one by linearity. Averaged over the 25 training *add-k* tasks, the label tokens contribute 96–99% of each main head’s task signal (Figure 15 in Appendix F.1), while input, separator, and query tokens contribute almost nothing.

Finally, we ask *what* information these tokens carry: for each task $k \in \{1, \dots, 30\}$, we measure the inner product $\langle W_h O_h V_h z_t, \tilde{h}_k / \|\tilde{h}_k\| \rangle$ between the extracted information and the unit-normalized projected head vector of §4.1. At each label token y_i of a random mixed- k prompt, this inner product peaks at $k = y_i - x_i$ (Figure 16a, Appendix F.1): the model extracts $y_i - x_i$ from each demonstration $x_i \rightarrow y_i$.

6 Related Work

Our work builds on interpretability research that aims to uncover circuits and internal computations of language models. Many studies focus on synthetic tasks or small models (Nanda et al., 2023; Bietti et al., 2023; Reddy, 2024; Singh et al., 2024). For large pretrained LLMs, some provide coarser-grained analyses (Todd et al., 2024; Olsson et al., 2022; Hendel et al., 2023), while others target particular model families and task classes for finer-grained insights (Hanna et al., 2023; Feng & Steinhardt, 2024; Wu et al., 2023; Zhou et al., 2024; Panickssery et al., 2024). Our work follows the latter, providing a deeper account of ICL mechanisms than prior studies.

Interpreting arithmetic tasks. A recent line of work examines how LLMs perform arithmetic (Stolfo et al., 2023; Hanna et al., 2023; Nikankin et al., 2025; Maltoni & Ferrara, 2024), and in particular addition (Nanda et al., 2023; Zhong et al., 2023; Zhou et al., 2024; Kantamneni & Tegmark, 2025). Zhou et al. (2024) find that pre-trained LLMs perform addition using Fourier features, and Kantamneni & Tegmark (2025) find that mid-sized LLMs compute addition using a “clock” algorithm via a helix representation of numbers. Unlike prior work, we analyze addition in the ICL setting for LLMs, and interestingly we find similar representational structures to them.

Interpreting in-context learning. Researchers have constructed detailed models of ICL for small transformers in supervised learning (Garg et al., 2022; Akyürek et al., 2023; Zhang et al., 2024; Li et al., 2023; Wu et al., 2024), as well as more complex settings (von Oswald et al., 2023; Bai et al., 2023; Bietti et al., 2023; Reddy, 2024; Guo et al., 2024; Nichani et al., 2024). For large pretrained models, ICL performance has been attributed to either induction heads (Olsson et al., 2022; Singh et al., 2024; Crosbie & Shutova, 2025; Bansal et al., 2023) or function vector (FV) heads (Todd et al., 2024; Hendel et al., 2023). Yin & Steinhardt (2025) finds that few-shot ICL performance depends primarily on FV heads. Motivated by this, we study FV heads in detail, introducing a novel optimization method that achieves better performance than Todd et al. (2024). The *add-k* tasks in our case study also share the same input domain, ensuring ICL prompts for different tasks differ only in the task information.

Causal analysis. Our methodological approach follows recent work on causal effects of model components via interventions on internal states (Vig et al., 2020; Geiger et al., 2021), drawing inspiration from causal mediation analysis (Todd et al., 2024), activation patching (Meng et al., 2022), and causal scrubbing (Chan et al., 2022).

7 Discussion

We provided a detailed mechanistic analysis of in-context learning for addition tasks. We found that a small number of attention heads operating in low-dimensional subspaces can extract, represent, and aggregate ICL task information in structured and interpretable ways. Our analysis proceeded in five steps: (1) sparse optimization to identify important heads (§3.1); (2) ablation to localize to a smaller subset (§3.2); (3) PCA to find low-dimensional subspaces (§4.1); (4) rotation and decomposition revealing periodic unit-digit and magnitude subspaces (§4.2.4.3); and (5) algebraic tracing connecting aggregation subspaces at the final token to extraction subspaces at earlier y_i tokens (§5). Most steps scale easily; future work could automate the qualitative subspace examination using AI.

Beyond automation, our analysis raises two further directions. First, *within* a task family, how do the identified heads and their subspaces function individually and together? Second, *across* task families, are the important heads and subspaces shared or transferable? Appendix B.1 offers initial evidence of sharing at the head level—one head, (15, 1), is a main head in every family we study—but whether the corresponding subspaces coincide, and whether a subspace identified for one family can steer another, remain open questions.

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A Overview Figure

Task: add-12 (e.g., $3 + 12 = 15$)

Prompt: “3 → 15 # 7 → 19 # 11 → 23 # 15 → 27 # 28 → 40 # 9 →”

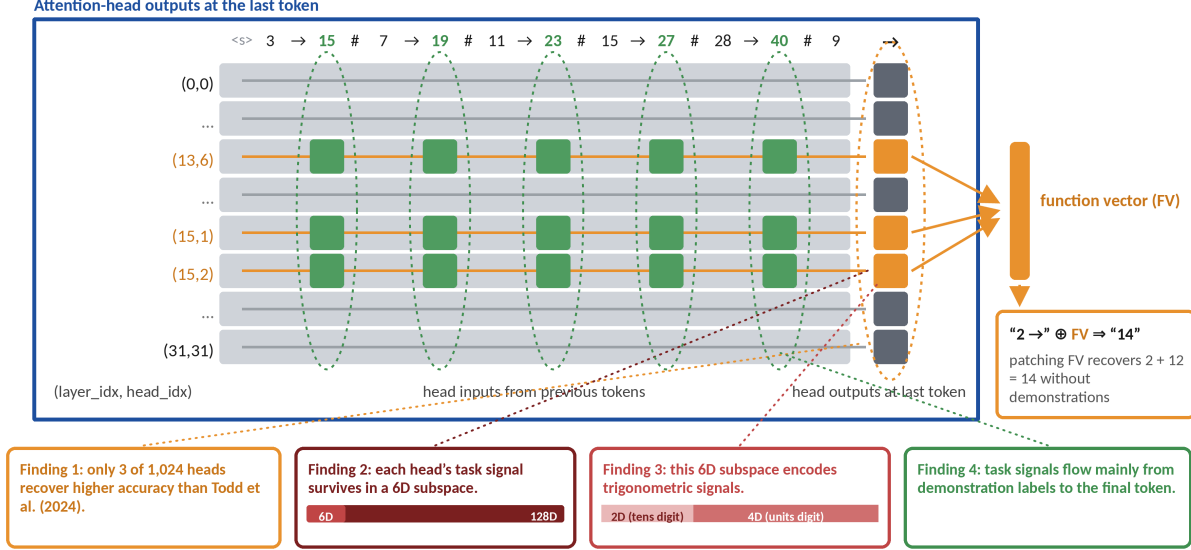


Figure 3: Key findings for Llama-3-8B-instruct, illustrated on an example five-shot *add-12* prompt: (1) only three of the 1024 attention heads carry most of the *add-k* task signal (§3); (2) each head’s task signal survives in a six-dimensional subspace of its outputs (§4.1); (3) that subspace encodes trigonometric functions of k and factors into a four-dimensional unit-digit part and a two-dimensional magnitude (tens-digit) part (§4.2, §4.3); and (4) the task signal flows mainly from the demonstration label tokens to the final token (§5), where it aggregates into a function vector (FV) that recovers the behavior when patched into a zero-shot prompt.

B Generality of the Localization Pipeline

This appendix applies the full pipeline of §3 — significant-head selection, recovery-positive refinement, main-head selection, and the subspace projection of §4.1 — to other task families (Table 3), models (Table 4), prompt formats (Table 6), and numbers of demonstrations (Table 7). In every variant that trains a coefficient matrix, significant heads are selected by the largest-gap rule of §3.1; the n -shot variants reuse the canonical five-shot selection.

Setup. For every variant, the family’s tasks are split into training tasks and 5 held-out tasks (per-family task counts in Table 2); coefficient-matrix training and all head selection see only the training tasks, and every clean and intervention accuracy is measured on the held-out tasks (100 queries each, i.e. 500 evaluations per number). Each entry reports the mean with a 95% bootstrap confidence interval (2,000 resamples) beneath it. The rows are: **clean** (n -shot accuracy without intervention); **corrupted** (zero-shot accuracy with every significant head mean-ablated; this baseline comes from the recovery scan and is measured on the training-task scan prompts); the **intervention accuracies** of the significant, recovery-positive, and main head sets (zero-shot prompts; the function vector is the unit-coefficient sum of the row’s head set, with the remaining significant heads mean-ablated), annotated with (heads used / total heads); **projected main heads** (each main head first projected onto its top principal components at 95% explained variance, annotated with PC dimensions kept / total head-output dimensions = $128 \times \#$ main heads); and the **AIE baseline** (function vector from the top- k heads ranked by a paper-faithful re-implementation of Todd et al. (2024)’s average indirect effect, at the small top- k budget of their paper: $k = 20$ for Llama-3-8B and Phi-4, $k = 16$ for Qwen2.5-7B — $\approx 2\%$ of heads for Llama-3 and Qwen). Non-arrow prompt formats use per-format retrained coefficient matrices; the n -shot variants reuse the canonical five-shot matrix and only re-run the recovery scan (20 queries per training task for $n \neq 5$). All accuracies use the exact-token-match scorer. In the tables, intervention accuracies and head budgets of our selected head sets are typeset in **red** and the AIE baseline in **blue**.

B.1 Task families

Table 2 specifies the five task families. Each arithmetic task fixes a constant k and applies a fixed rule to an input integer $x \in [1, 100]$, with 100 datapoints per task: the addition family is *add- k* with $k \in [1, 30]$ (§2.1), the subtraction family is *add- k* with $k \in [-30, 0]$ (including the identity task $k = 0$), and the multiplication family is *mul- k* ($y = kx$) with $k \in [1, 30]$ (including the identity task $k = 1$). The abstractive and extractive families use the task datasets of Todd et al. (2024). An abstractive task maps its input to a semantically associated output that need not appear in the input: antonym and synonym; English–French and English–Spanish translation; country-capital, country-currency, landmark-country, park-country, and national-parks; person-occupation, person-sport, and person-instrument; product-company; present-past; singular-plural; sentiment; next-item and previous-item; next-capital-letter; word-length; and six capitalization/lowercasing manipulations (26 tasks in total). An extractive task outputs a span of its input. Most are word-list tasks: given a list of words, choose the first, middle, or last item; the alphabetically first or last item; or the item of the first-named category (adjective-vs-verb, verb-vs-adjective, animal-vs-object, color-vs-animal, concept-vs-object, object-vs-concept, and fruit-vs-animal — e.g., the animal in animal-vs-object); each of these 12 rules is a separate task for lists of 3 and of 5 words. The remaining three tasks are CoNLL-2003 named-entity extraction: given a news sentence or fragment, output the (possibly multi-word) person, location, or organization it mentions (27 tasks in total).

Beyond addition, we run the pipeline on subtraction, multiplication, and the extractive and abstractive families of Todd et al. (2024), all on Llama-3-8B-instruct (Table 3). The localization structure is shared but family-dependent: one head, (15, 1), is a main head in every family; the subtraction main heads are exactly the addition trio plus (15, 17); multiplication is led by (13, 6) (with (15, 1), (15, 17), (15, 28), and (13, 21)); and the semantic families are genuinely more distributed — 11 (extractive) and 10 (abstractive) main heads with a broader layer spread — while still led by (15, 1).

Family	Task rule	Tasks (train + held-out)	Example task: datapoint
Addition	$y = x + k, k \in [1, 30]$	30 (25 + 5)	add-7: 3 $\rightarrow$ 10
Subtraction	$y = x + k, k \in [-30, 0]$	31 (26 + 5)	add-(-7): 3 $\rightarrow$ -4
Multiplication	$y = kx, k \in [1, 30]$	30 (25 + 5)	mul-4: 3 $\rightarrow$ 12
Abstractive	semantic association	26 (21 + 5)	product-company: iPhone 5 $\rightarrow$ Apple
Extractive	copy an input span	27 (22 + 5)	animal-vs-object: bear, ukulele, blanket $\rightarrow$ bear

Table 2: The five task families. Arithmetic inputs range over $x \in [1, 100]$ (100 datapoints per task); the abstractive and extractive task datasets come from [Todd et al. \(2024\)](#) and vary in size. Within each family, the split into training tasks and 5 held-out tasks is fixed once and reused across all models, prompt formats, and shot counts; every clean and intervention accuracy is measured on the held-out tasks. Examples are verbatim dataset entries.

	Subtraction	Multiplication	Extractive	Abstractive
Clean accuracy	0.526 [0.484, 0.572]	0.348 [0.308, 0.390]	0.810 [0.776, 0.846]	0.422 [0.378, 0.466]
Corrupted accuracy	0.020 [0.015, 0.025]	0.030 [0.024, 0.037]	0.309 [0.289, 0.328]	0.023 [0.017, 0.030]
Significant heads	0.330 (28/1024) [0.290, 0.372]	0.226 (37/1024) [0.190, 0.264]	0.506 (15/1024) [0.460, 0.548]	0.300 (23/1024) [0.260, 0.342]
Recovery-positive heads	0.326 (19/1024) [0.286, 0.368]	0.134 (13/1024) [0.104, 0.162]	0.596 (13/1024) [0.554, 0.640]	0.290 (19/1024) [0.250, 0.330]
Main heads	0.290 (4/1024) [0.252, 0.330]	0.080 (5/1024) [0.058, 0.104]	0.564 (11/1024) [0.522, 0.608]	0.248 (10/1024) [0.210, 0.286]
Projected main heads	0.282 (28/512) [0.244, 0.320]	0.076 (40/640) [0.054, 0.100]	0.538 (109/1408) [0.494, 0.582]	0.218 (130/1280) [0.182, 0.256]
AIE top- k heads	0.002 ($k=20$) [0.000, 0.006]	0.000 ($k=20$) [0.000, 0.000]	0.448 ($k=20$) [0.406, 0.494]	0.168 ($k=20$) [0.136, 0.200]

Table 3: Generalization across task families (Llama-3-8B-instruct, arrow format, five-shot). Rows as defined in the setup paragraph; brackets are 95% bootstrap confidence intervals. Parentheses report the budget of each intervention: (number of heads used / total heads in the model) for the significant, recovery-positive, and main rows; (PC dimensions kept / total dimensions of the main heads’ outputs, 128 per head) for the projected row; and the top- k head budget for the AIE row. The extractive corrupted floor reflects the multiple-choice nature of that family (chance ≈ 0.31).

B.2 Models

Table 4 applies the pipeline to *add-k* on Phi-4-14B and Qwen2.5-7B. Phi-4 localizes to three main heads (18, 12), (18, 14), (17, 23) in layers 17–18 (of 40 layers and 1600 heads) and mirrors the Llama-3 family structure: (18, 12) is a main head for every Phi-4 task family, and its subtraction main heads are exactly its addition main heads plus (17, 22). Table 5 reports the Phi-4 task-family results, the analogue of Table 3; the per-family main-head sets are subtraction $\{(18, 12), (18, 14), (17, 22), (17, 23)\}$, multiplication $\{(18, 12), (11, 9), (18, 14), (24, 2)\}$, extractive $\{(24, 31), (19, 4), (18, 12), (21, 25), (17, 22), (17, 23)\}$, and abstractive (14 heads, again including (18, 12)). Note that the Phi-4 subtraction interventions from our selected head sets sit *above* the clean accuracy. Qwen2.5-7B selects exactly the layer-21 quartet (21, 0), (21, 5), (21, 4), (21, 2). Qwen’s unit-coefficient composites carry a metric caveat: its tokenizer splits numbers into digits, which the exact-token-match scorer punishes; its trained-coefficient function vector reaches 0.460.

	Llama-3-8B-instruct	Phi-4-14B	Qwen2.5-7B
Clean accuracy	0.870 [0.838, 0.898]	0.928 [0.904, 0.950]	0.896 [0.870, 0.922]
Corrupted accuracy	0.037 [0.030, 0.044]	0.031 [0.024, 0.038]	0.032 [0.026, 0.039]
Significant heads	0.798 (33/1024) [0.760, 0.832]	0.908 (23/1600) [0.882, 0.932]	0.368 (230/784) [0.326, 0.410]
Recovery-positive heads	0.774 (13/1024) [0.736, 0.810]	0.910 (13/1600) [0.884, 0.934]	0.358 (18/784) [0.314, 0.400]
Main heads	0.680 (3/1024) [0.640, 0.720]	0.876 (3/1600) [0.848, 0.904]	0.270 (4/784) [0.232, 0.310]
Projected main heads	0.636 (18/384) [0.592, 0.678]	0.870 (22/384) [0.842, 0.898]	0.252 (28/512) [0.216, 0.290]
AIE top- k heads	0.124 ($k=20$) [0.096, 0.152]	0.728 ($k=20$) [0.690, 0.766]	0.136 ($k=16$) [0.104, 0.168]

Table 4: Generalization across models (*add-k*, arrow format, five-shot). Rows as defined in the setup paragraph; brackets are 95% bootstrap confidence intervals. Parentheses report the budget of each intervention: (number of heads used / total heads in the model) for the significant, recovery-positive, and main rows; (PC dimensions kept / total dimensions of the main heads’ outputs, 128 per head) for the projected row; and the top- k head budget for the AIE row.

	Subtraction	Multiplication	Extractive	Abstractive
Clean accuracy	0.658 [0.618, 0.700]	0.426 [0.382, 0.468]	0.768 [0.732, 0.808]	0.442 [0.398, 0.486]
Corrupted accuracy	0.036 [0.029, 0.043]	0.030 [0.024, 0.036]	0.261 [0.243, 0.279]	0.033 [0.026, 0.040]
Significant heads	0.754 (37/1600) [0.718, 0.792]	0.438 (36/1600) [0.392, 0.482]	0.412 (18/1600) [0.370, 0.456]	0.326 (31/1600) [0.286, 0.366]
Recovery-positive heads	0.776 (13/1600) [0.740, 0.814]	0.438 (24/1600) [0.392, 0.482]	0.416 (16/1600) [0.372, 0.460]	0.324 (29/1600) [0.284, 0.364]
Main heads	0.754 (4/1600) [0.718, 0.792]	0.128 (4/1600) [0.102, 0.158]	0.402 (6/1600) [0.360, 0.444]	0.234 (14/1600) [0.198, 0.272]
Projected main heads	0.754 (30/512) [0.718, 0.792]	0.120 (38/512) [0.094, 0.148]	0.390 (63/768) [0.348, 0.432]	0.182 (187/1792) [0.148, 0.216]
AIE top- k heads	0.038 ($k=20$) [0.022, 0.056]	0.004 ($k=20$) [0.000, 0.010]	0.344 ($k=20$) [0.302, 0.388]	0.180 ($k=20$) [0.146, 0.214]

Table 5: Generalization across task families for Phi-4-14B (arrow format, five-shot). Rows as defined in the setup paragraph; brackets are 95% bootstrap confidence intervals. Parentheses report the budget of each intervention: (number of heads used / total heads in the model) for the significant, recovery-positive, and main rows; (PC dimensions kept / total dimensions of the main heads’ outputs, 128 per head) for the projected row; and the top- k head budget for the AIE row.

B.3 Prompt formats

Table 6 varies the prompt format on Llama-3-8B *add-k*, with a separately retrained coefficient matrix per format. Clean accuracy varies substantially across formats and the significant sets differ (24–46 heads), but the localization is structurally stable: **every format selects the three main heads** (15, 2), (15, 1), (13, 6) (formats qa and ab additionally select (15, 17), giving four main heads there).

	arrow (orig.)	qa	io	ab	textlabel	fx
Clean accuracy	0.870 [0.838, 0.898]	0.418 [0.372, 0.462]	0.532 [0.490, 0.574]	0.700 [0.660, 0.738]	0.832 [0.798, 0.864]	0.348 [0.306, 0.390]
Corrupted accuracy	0.037 [0.030, 0.044]	0.018 [0.013, 0.023]	0.034 [0.026, 0.041]	0.030 [0.024, 0.037]	0.018 [0.013, 0.024]	0.032 [0.025, 0.039]
Significant heads	0.798 (33/1024) [0.760, 0.832]	0.498 (28/1024) [0.454, 0.542]	0.470 (46/1024) [0.426, 0.512]	0.736 (38/1024) [0.696, 0.772]	0.582 (24/1024) [0.540, 0.624]	0.270 (32/1024) [0.230, 0.310]
Recovery-positive heads	0.774 (13/1024) [0.736, 0.810]	0.492 (26/1024) [0.448, 0.536]	0.440 (9/1024) [0.396, 0.482]	0.726 (15/1024) [0.686, 0.762]	0.558 (21/1024) [0.516, 0.598]	0.270 (11/1024) [0.230, 0.308]
Main heads	0.680 (3/1024) [0.640, 0.720]	0.432 (4/1024) [0.388, 0.476]	0.334 (3/1024) [0.294, 0.374]	0.650 (4/1024) [0.608, 0.690]	0.496 (3/1024) [0.452, 0.538]	0.190 (3/1024) [0.156, 0.224]
Projected main heads	0.636 (18/384) [0.592, 0.678]	0.410 (24/512) [0.366, 0.456]	0.306 (18/384) [0.266, 0.348]	0.614 (24/512) [0.570, 0.656]	0.464 (18/384) [0.418, 0.506]	0.184 (18/384) [0.150, 0.218]
AIE top- k heads ($k=20$)	0.124 [0.096, 0.152]	0.126 [0.096, 0.156]	0.078 [0.056, 0.102]	0.196 [0.160, 0.230]	0.104 [0.078, 0.130]	0.064 [0.042, 0.086]

Table 6: Generalization across prompt formats (Llama-3-8B-instruct *add-k*, five-shot; per-format retrained coefficient matrices). Rows as defined in the setup paragraph; brackets are 95% bootstrap confidence intervals. Parentheses report the budget of each intervention: (number of heads used / total heads in the model) for the significant, recovery-positive, and main rows; (PC dimensions kept / total dimensions of the main heads’ outputs, 128 per head) for the projected row; and the top- k head budget for the AIE row.

B.4 Number of demonstrations

Table 7 varies the number of demonstrations n from 1 to 10 while reusing the canonical five-shot coefficient matrix. Clean accuracy rises from 0.256 to 0.944 (saturating around $n \approx 7$), while the intervention accuracies plateau from $n \approx 4$. The localization is again structurally stable: **every n selects the same three main heads**. The AIE baseline, by contrast, depends strongly on the demonstration budget (it ranks heads by corrupting demonstration labels): at $n=1$ its top-33 list omits (15, 2) and (13, 6) entirely, and its top-20 accuracy climbs from 0.014 to ~ 0.3 only at large n .

n	1	2	3	4	5 (orig.)	6	7	8	9	10
Clean accuracy	0.256 [0.216, 0.294]	0.502 [0.458, 0.544]	0.700 [0.658, 0.740]	0.816 [0.780, 0.850]	0.870 [0.838, 0.898]	0.898 [0.870, 0.924]	0.932 [0.908, 0.954]	0.944 [0.924, 0.962]	0.940 [0.918, 0.960]	0.944 [0.924, 0.962]
Corrupted accuracy	0.046 [0.030, 0.066]	0.034 [0.020, 0.050]	0.028 [0.014, 0.044]	0.032 [0.018, 0.048]	0.037 [0.030, 0.044]	0.032 [0.018, 0.048]	0.034 [0.020, 0.050]	0.032 [0.018, 0.048]	0.032 [0.018, 0.048]	0.032 [0.018, 0.048]
Significant heads (33/1024)	0.188 [0.154, 0.222]	0.586 [0.544, 0.630]	0.752 [0.714, 0.790]	0.782 [0.746, 0.818]	0.798 [0.760, 0.832]	0.774 [0.736, 0.808]	0.772 [0.736, 0.808]	0.770 [0.734, 0.806]	0.772 [0.736, 0.806]	0.754 [0.716, 0.790]
Recovery-positive heads	0.184 (6) [0.150, 0.218]	0.534 (8) [0.490, 0.578]	0.700 (10) [0.658, 0.740]	0.776 (10) [0.738, 0.812]	0.774 (13) [0.736, 0.810]	0.756 (11) [0.716, 0.792]	0.746 (10) [0.706, 0.782]	0.750 (10) [0.712, 0.786]	0.748 (10) [0.710, 0.784]	0.740 (10) [0.702, 0.778]
Main heads (3/1024)	0.156 [0.122, 0.186]	0.452 [0.408, 0.496]	0.626 [0.582, 0.668]	0.674 [0.632, 0.716]	0.680 [0.640, 0.720]	0.664 [0.622, 0.704]	0.652 [0.608, 0.694]	0.652 [0.608, 0.692]	0.644 [0.600, 0.684]	0.628 [0.584, 0.670]
Projected main heads (18/384)	0.156 [0.122, 0.186]	0.410 [0.366, 0.452]	0.592 [0.548, 0.636]	0.638 [0.594, 0.680]	0.636 [0.592, 0.678]	0.632 [0.588, 0.674]	0.618 [0.572, 0.660]	0.616 [0.570, 0.658]	0.620 [0.574, 0.662]	0.620 [0.574, 0.662]
AIE top- k heads ($k=20$)	0.014 [0.004, 0.026]	0.042 [0.026, 0.060]	0.072 [0.050, 0.096]	0.074 [0.052, 0.098]	0.124 [0.096, 0.152]	0.312 [0.270, 0.350]	0.186 [0.152, 0.220]	0.318 [0.276, 0.356]	0.172 [0.138, 0.204]	0.316 [0.274, 0.354]

Table 7: Generalization across the number of demonstrations n (Llama-3-8B-instruct *add-k*, arrow format; the five-shot coefficient matrix is reused, so the significant set is fixed at 33 heads). Recovery-positive counts are per- n (out of 1024). Recovery scans for $n \neq 5$ use 20 queries per training task. Brackets are 95% bootstrap confidence intervals. Parenthetical row-label annotations report the budget of each intervention: (number of heads used / total heads in the model) for the significant and main rows; (PC dimensions kept / total dimensions of the main heads’ outputs, 128 per head) for the projected row; and the top- k head budget for the AIE row.

B.5 Comparison with the AIE baseline

The AIE rows above evaluate Todd et al. (2024)’s head selection at the small top- k budget of their paper. The comparison is not driven by the budget: at matched k equal to our significant-set size, the AIE top- k set also trails our significant set in every variant — on

canonical *add-k*, 0.414 [0.372, 0.456] at $k=33$ versus our 0.798 [0.760, 0.832]. In **every** task family, model, prompt format, and shot count above, our 3–14 main heads achieve higher intervention accuracy than the AIE top- k set while using fewer heads; the pattern also reproduces on the Phi-4 non-addition families (Table 5). We note one nuance in fairness to AIE: its per-head *ranking* is informative — it places our main heads at or near the top of the per-head list in essentially every variant — but its top- k *sets* underperform because the tail of the ranking does not sum into an effective function vector.

B.6 Subspace dimensionality across variants

The six-dimensional finding of §4.1 generalizes as low-dimensional structure with family-dependent dimensionality. The number of principal components needed to reach 95% of task variance for main heads is 6 per head on Llama-3 addition (every format and every n), 7 on subtraction, 6–9 on multiplication, 7–8 on Phi-4 addition, and 5–8 on Qwen2.5 addition, rising to 6–11 (extractive) and 11–15 (abstractive) for the semantic families. Causal sufficiency is universal: as the projected rows of Tables 3–7 show, replacing every main head’s vector by its projection onto these subspaces costs at most 0.052 accuracy in any displayed cell (0.058 across the full grid including recovery-positive sets).

C Supplement for §4.1

We perform PCA on the 30 task vectors of each main head and plot the cumulative explained variance ratio against the number of PCs (Figure 4). Six directions explain 97% of the task variance for every Llama-3-8B-instruct main head (Figure 4a); the Phi-4 main heads reach 95% at 7–8 PCs and the Qwen-2.5-7B main heads at 5–8 PCs (Figures 4b and 4c).

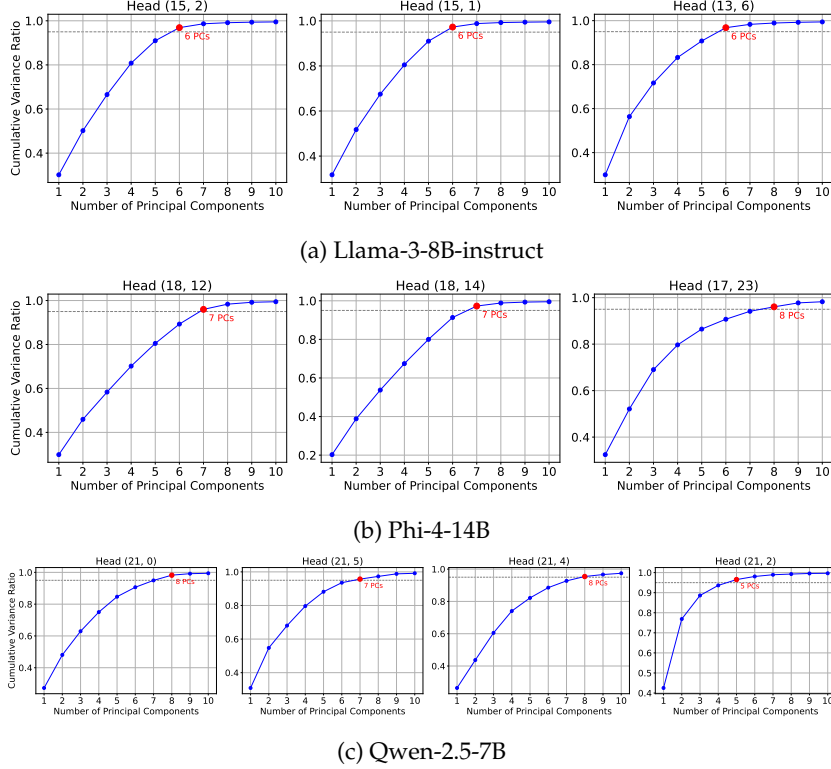


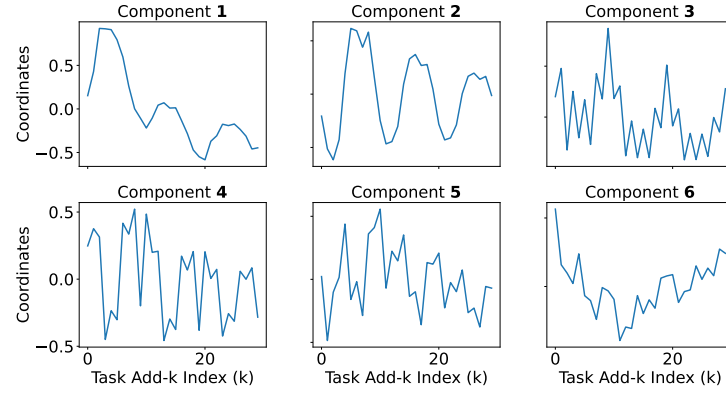
Figure 4: Cumulative explained variance ratio vs. number of PCs for each main head. The first six PCs capture 97% of the task variance for the Llama-3-8B-instruct main heads; the Phi-4 main heads reach 95% at 7–8 PCs and the Qwen-2.5-7B main heads at 5–8 PCs.

D Supplement for §4.2

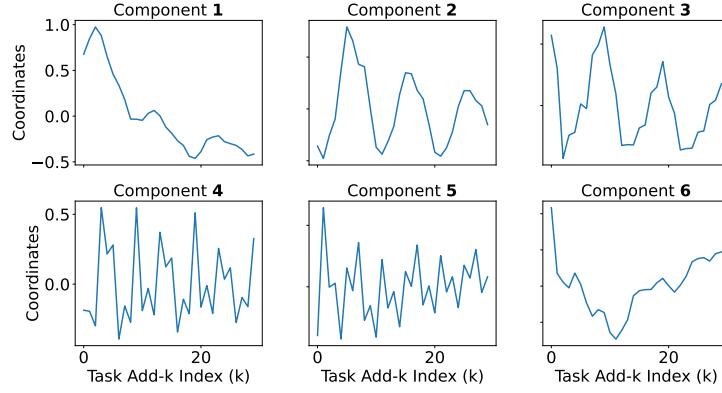
We first present additional figures for Llama-3-8B-Instruct, followed by results for Phi-4 and Qwen-2.5-7B. The periodic structure generalizes: all Phi-4 main heads fit the six trigonometric patterns well, while for Qwen-2.5 the periodic (unit-digit) patterns are fit well by heads (21,0) and (21,4) but poorly by (21,2) and (21,5), whose coordinates instead follow the two low-frequency (magnitude) directions and behave distinctly across the intervals $[1, 10]$, $[11, 20]$, and $[21, 30]$. We conjecture this difference arises from tokenization: Qwen encodes numbers digit by digit, leading to discontinuities across each 10-interval, whereas Llama-3 and Phi-4 represent every number in our range as a single token.

D.1 Additional figures for §4.2

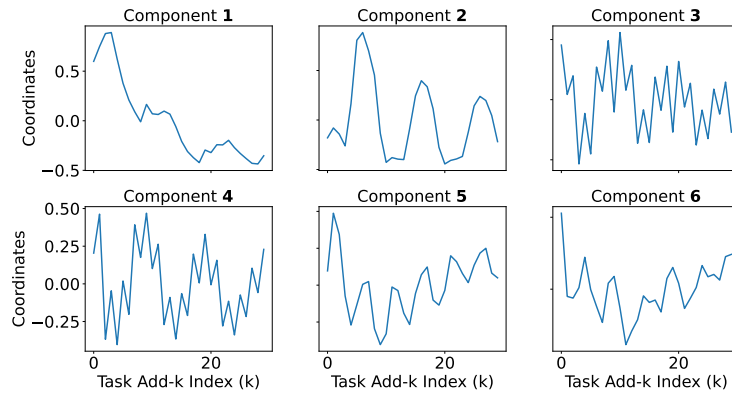
In §4.2, we found six trigonometric functions that can be linearly fitted by the coordinate functions. Here we first supplement the plot from which we observe the periodic pattern of the coordinate functions for the three heads (Figure 5) and then show the fitting functions for three heads (Figure 6).



(a) Head (15, 2)

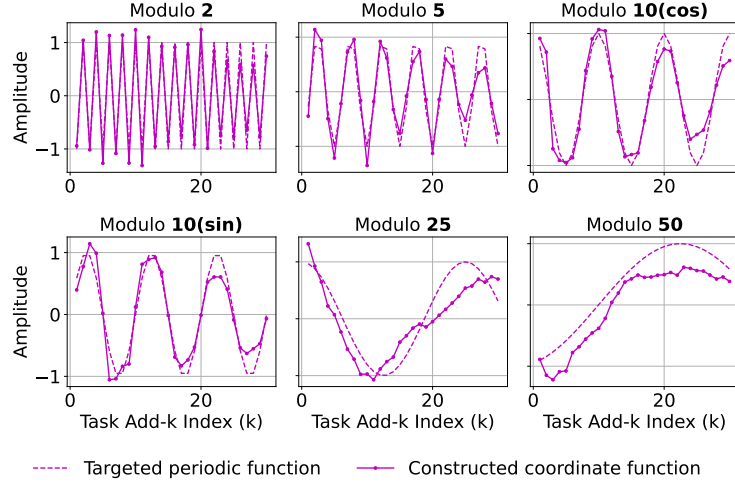


(b) Head (15, 1)

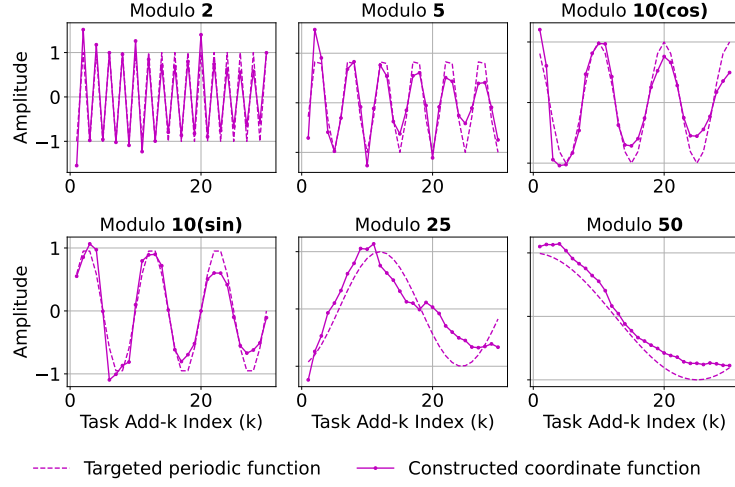


(c) Head (13, 6)

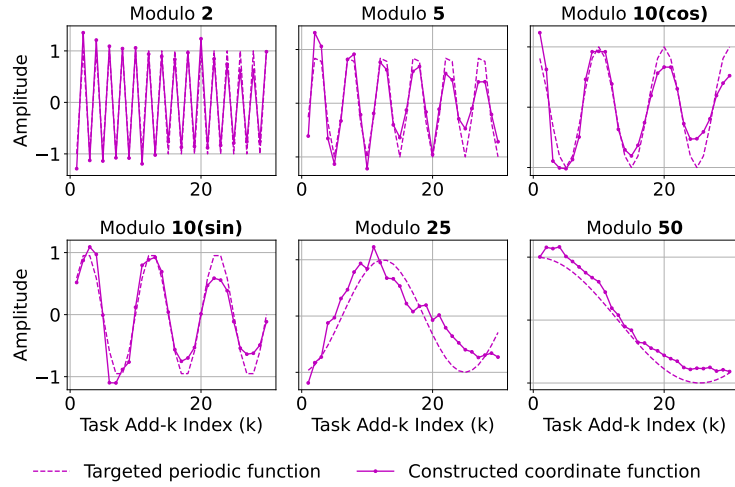
Figure 5: Coordinates of three heads' vectors (inner products with PCs) for the first six PCs across different add- k tasks on Llama-3-8B-instruct. Periodic patterns are visible in the first few PCs of each head.



(a) Head (15, 2)



(b) Head (15, 1)

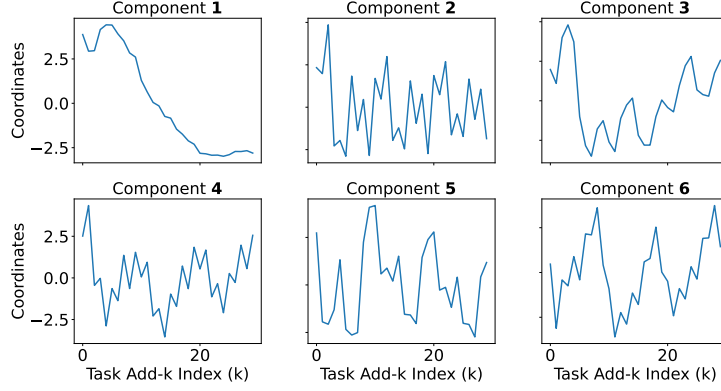


(c) Head (13, 6)

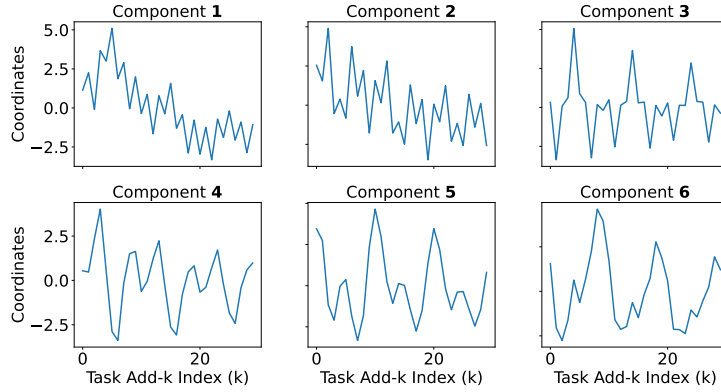
Figure 6: Coordinate functions of three heads can fit six trigonometric functions with periods 2, 5, 10, 10, 25, and 50 on Llama-3-8B-instruct.

D.2 Phi-4-14B for §4.2

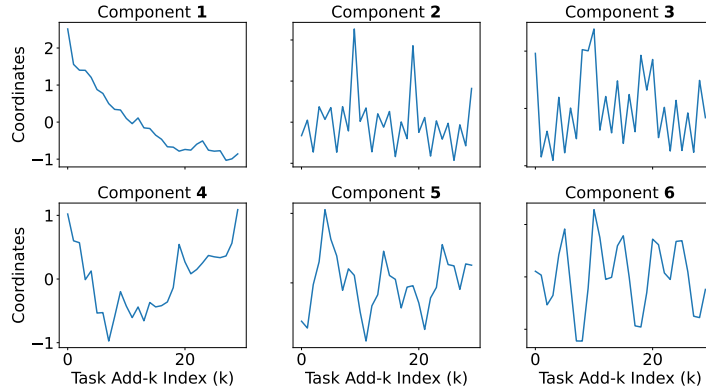
The Phi-4 main heads show similar periodic structure to Llama-3-8B-instruct: their PC coordinates are periodic in k (Figure 7) and fit the six trigonometric patterns well (Figure 8).



(a) Head (18, 12)

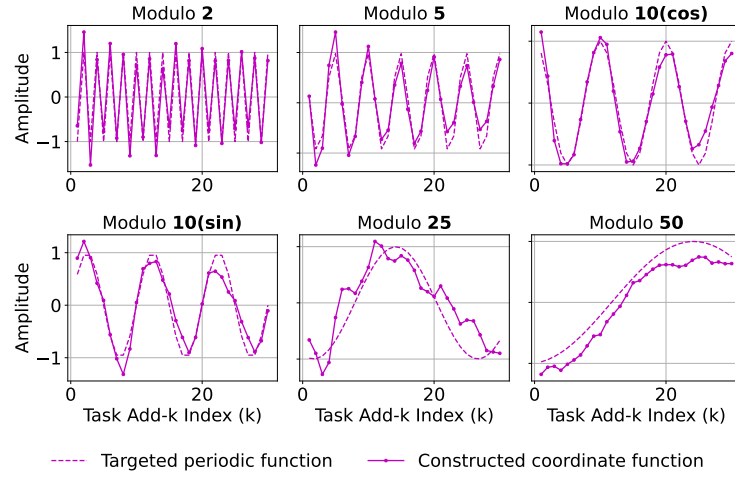


(b) Head (18, 14)

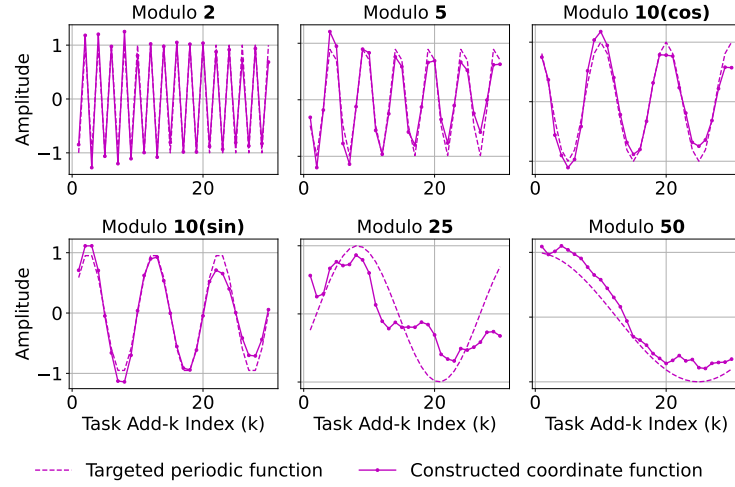


(c) Head (17, 23)

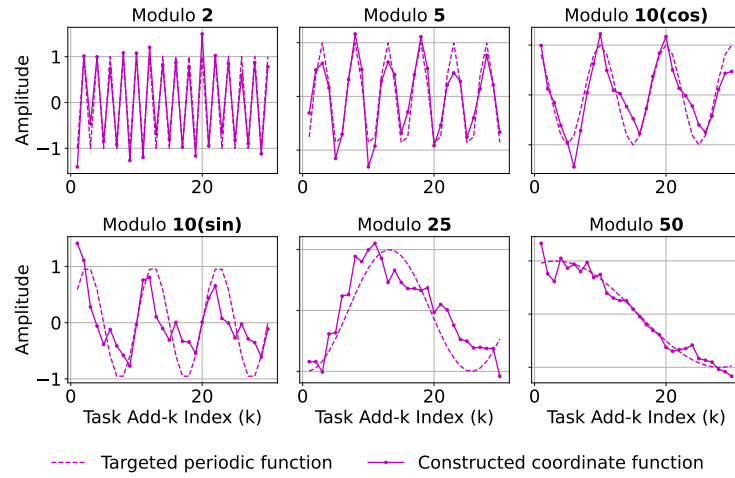
Figure 7: Coordinates of the three Phi-4 main heads' vectors (inner products with PCs) for the first six PCs across different add- k tasks. Periodic patterns are visible in the first few PCs of each head.



(a) Head (18, 12)



(b) Head (18, 14)



(c) Head (17, 23)

Figure 8: Coordinate functions of the three Phi-4 main heads fit six trigonometric functions with periods 2, 5, 10, 10, 25, and 50.

D.3 Qwen-2.5-7B for §4.2

Heads (21, 0) and (21, 4) show periodic patterns similar to the Llama-3 heads, while heads (21, 2) and (21, 5) do not encode the unit-digit periodicity; their coordinates instead follow the low-frequency magnitude directions and behave distinctly across the intervals $[1, 10]$, $[11, 20]$, and $[21, 30]$ (Figures 9 and 10). We conjecture this difference arises from tokenization: Qwen encodes numbers digit by digit, creating discontinuities across each 10-interval.

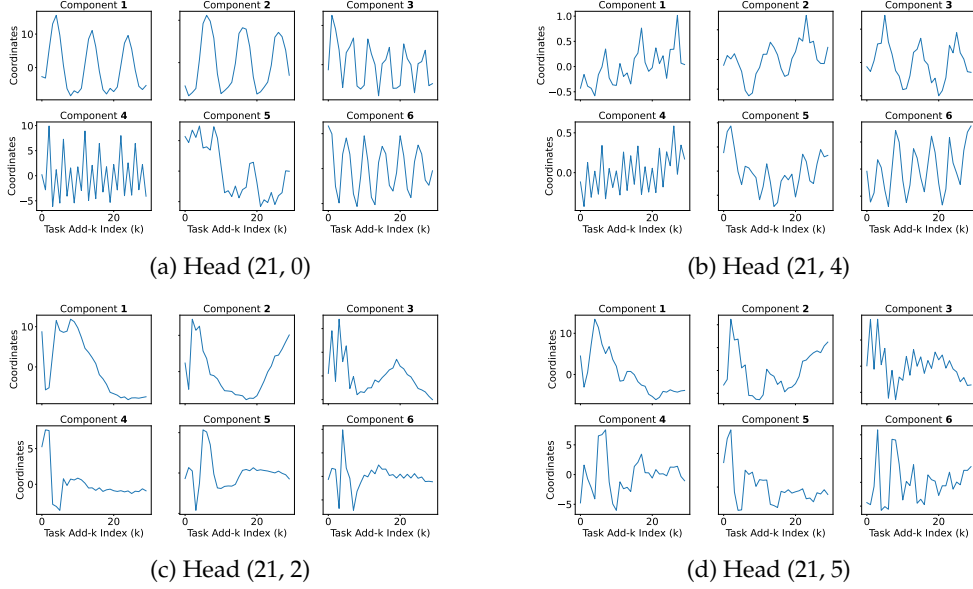


Figure 9: Coordinates of the four Qwen-2.5-7B main heads’ vectors (inner products with PCs) for the first six PCs across different add- k tasks. Heads (21,0) and (21,4) reveal periodic patterns, whereas heads (21,2) and (21,5) show discontinuous behaviors across the intervals $[1, 10]$, $[11, 20]$, and $[21, 30]$.

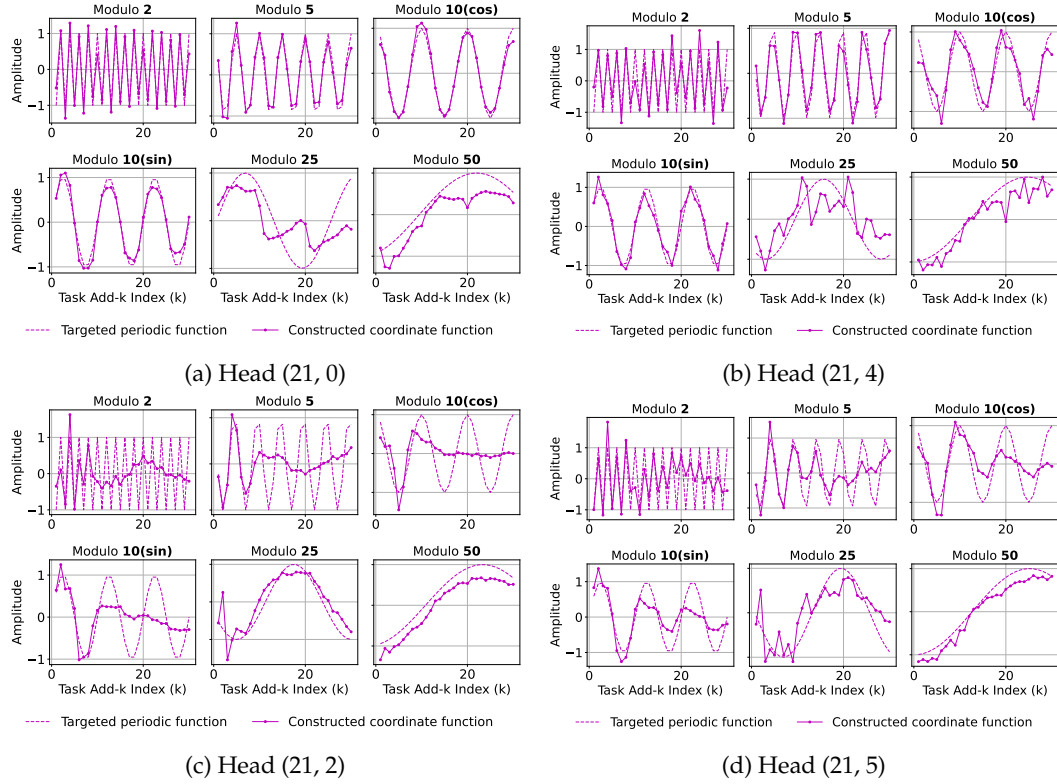


Figure 10: Six trigonometric functions with periods 2, 5, 10, 10, 25, and 50 fitted by the coordinate functions of the four Qwen-2.5-7B main heads. Heads (21, 0) and (21, 4) fit the periodic functions well, while heads (21, 2) and (21, 5) fit only the low-frequency directions, consistent with their non-periodic coordinate behaviors.

E Supplement for §4.3

Recall that the three hypotheses in §4.3 are as follows:

- (i) the feature direction corresponding to period two, which we call the “parity direction”, encodes the parity of k in the $add-k$ task;
- (ii) the subspace spanned by the feature directions with periods 2, 5, 10, which we call the “unit subspace”, encodes the unit digit of k ;
- (iii) the subspace spanned by the directions with periods 25, 50, which we call the “magnitude subspace”, encodes the coarse magnitude (i.e., the tens digit) of k .

We first show the experimental results validating hypothesis (ii) for head 1 (Figure 11), and then show analogous results for hypotheses (i) and (iii) and for the other two heads (Figures 12–14). Projecting out of the parity direction does not substantially increase errors for either the parity or the final answer across all tasks, perhaps because parity is easy to obtain (e.g., random choice leads to 0.5 accuracy).

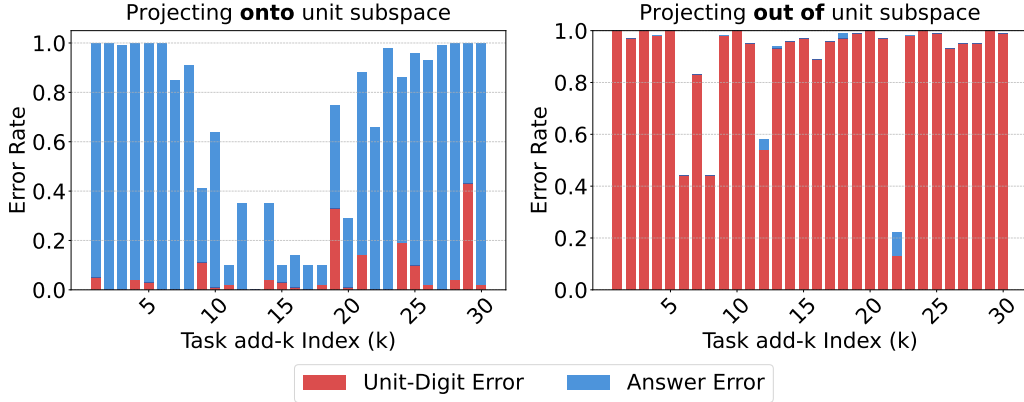


Figure 11: The error rates for the **unit digit** and the **final answer** across tasks when projecting head 1’s vectors **onto** (left) and **out of** (right) the “unit subspace”. Projecting onto the unit subspace yields a low unit-digit error rate even when the final-answer error remains high, while projecting out leads to high unit-digit error rates that almost fully account for final-answer errors. This confirms that the unit subspace encodes the unit-digit signal.

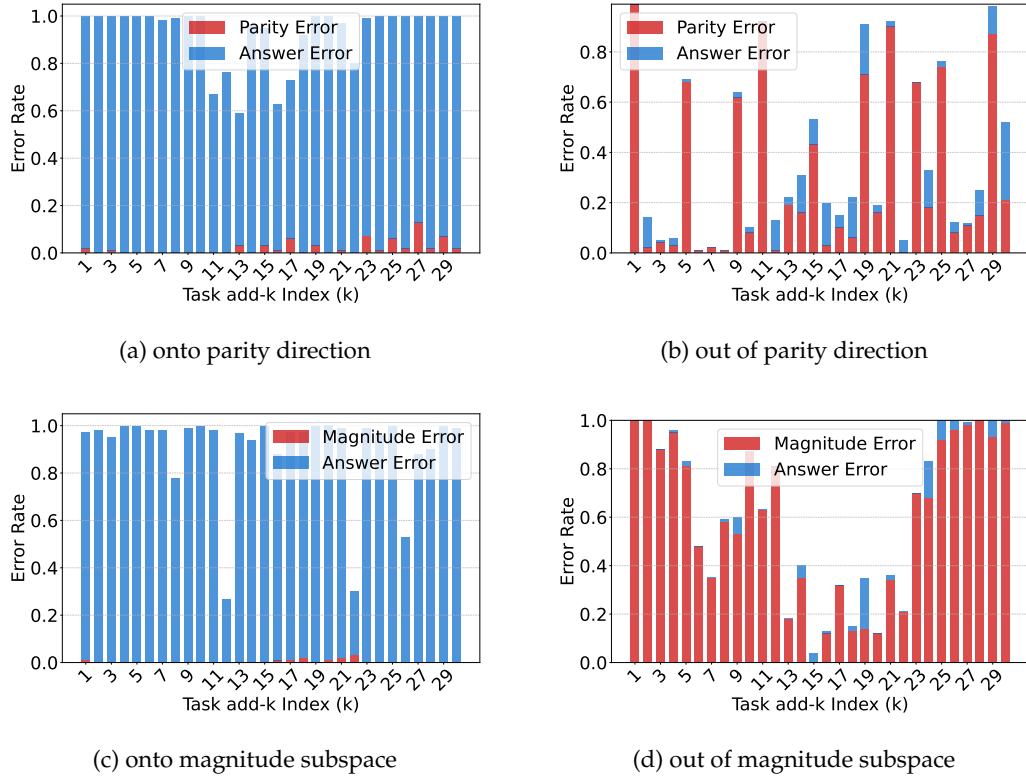
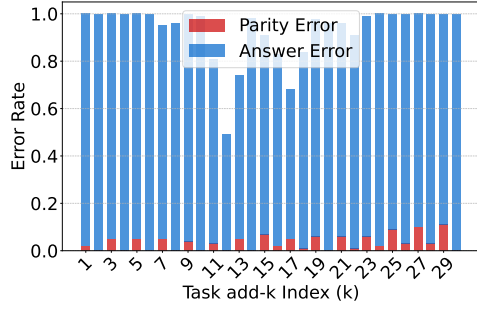
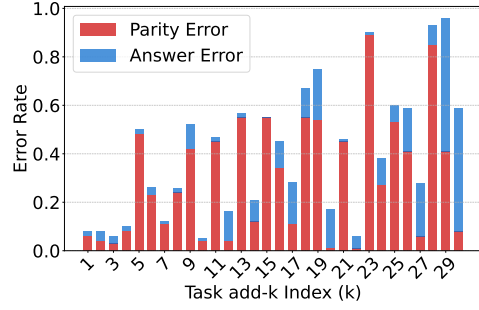


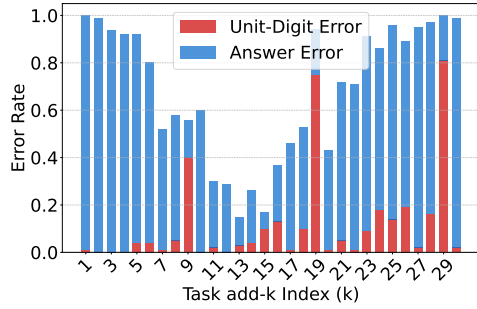
Figure 12: Validation of hypotheses (i) and (iii) for head 1. Each row shows results for the parity and magnitude subspaces (left: projection **onto**; right: projection **out of**). The unit-digit hypothesis (ii) is omitted here since it is already shown in Figure 11.



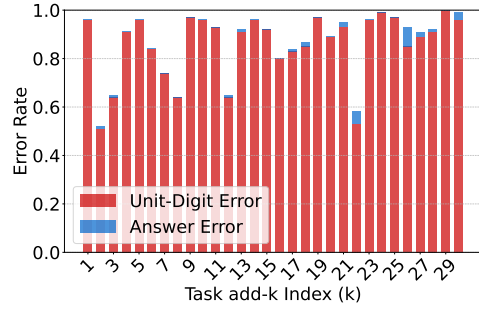
(a) Head 2 onto parity direction



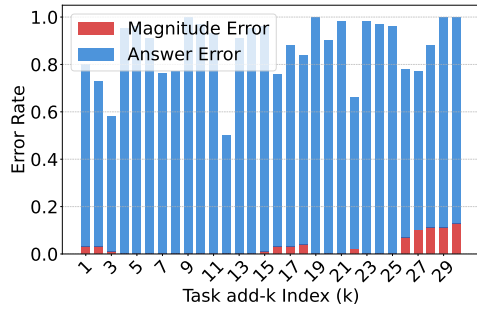
(b) Head 2 out of parity direction



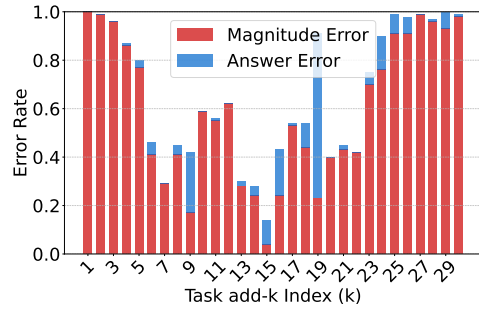
(c) Head 2 onto unit subspace



(d) Head 2 out of unit subspace

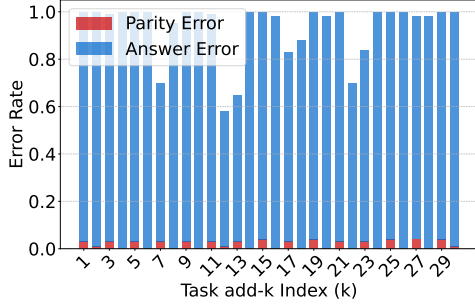


(e) Head 2 onto magnitude subspace



(f) Head 2 out of magnitude subspace

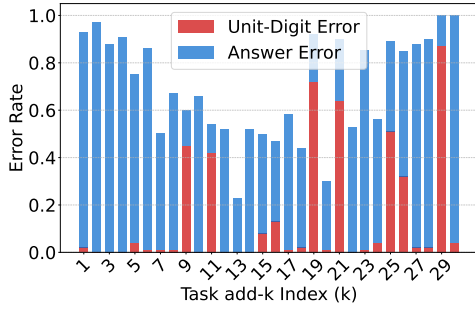
Figure 13: Validation of hypotheses (i)–(iii) for head 2. Each row shows the projection effects for the parity, unit, and magnitude subspaces respectively.



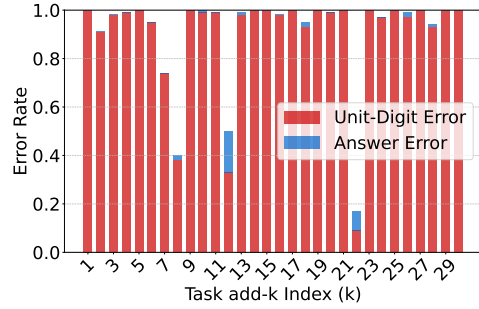
(a) Head 3 onto parity direction



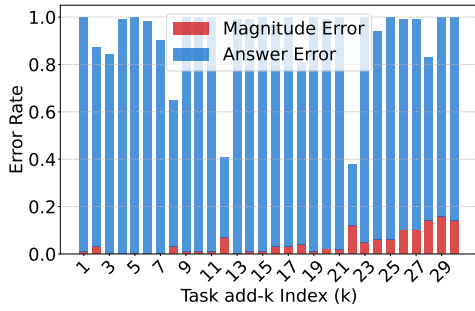
(b) Head 3 out of parity direction



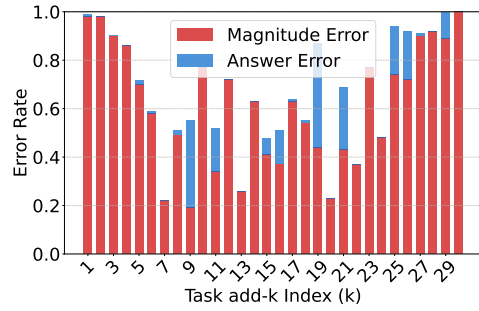
(c) Head 3 onto unit subspace



(d) Head 3 out of unit subspace



(e) Head 3 onto magnitude subspace



(f) Head 3 out of magnitude subspace

Figure 14: Validation of hypotheses (i)–(iii) for head 3. The evidence is slightly weaker than for heads 1 and 2, consistent with head 3’s lower intervened accuracy (§3.2).

F Supplement for §5.2

This appendix provides the figures supporting §5.2.

F.1 Additional figures for §5.2

First, Figure 15 reports the per-position decomposition of each main head’s task signal on standard five-shot prompts.

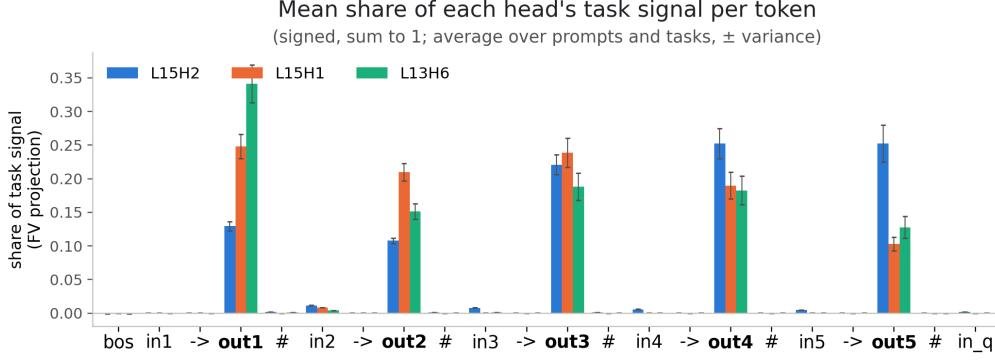


Figure 15: Mean share of each main head’s task signal contributed by each token position on five-shot *add-k* prompts (signed shares sum to one; averaged over prompts and the 25 training tasks; error bars show variance). The demonstration label tokens y_i (bold) carry essentially the entire task signal for all three main heads.

Next, we turn to the signals extracted from each token of a mixed- k ICL prompt (“3→9 # 15→17 # 62→78 # 20→24 # 11→30 # 35→”). All three heads behave similarly: the extracted signal is strongest at the y_i tokens, and its direction encodes $y_i - x_i$ (Figure 16).

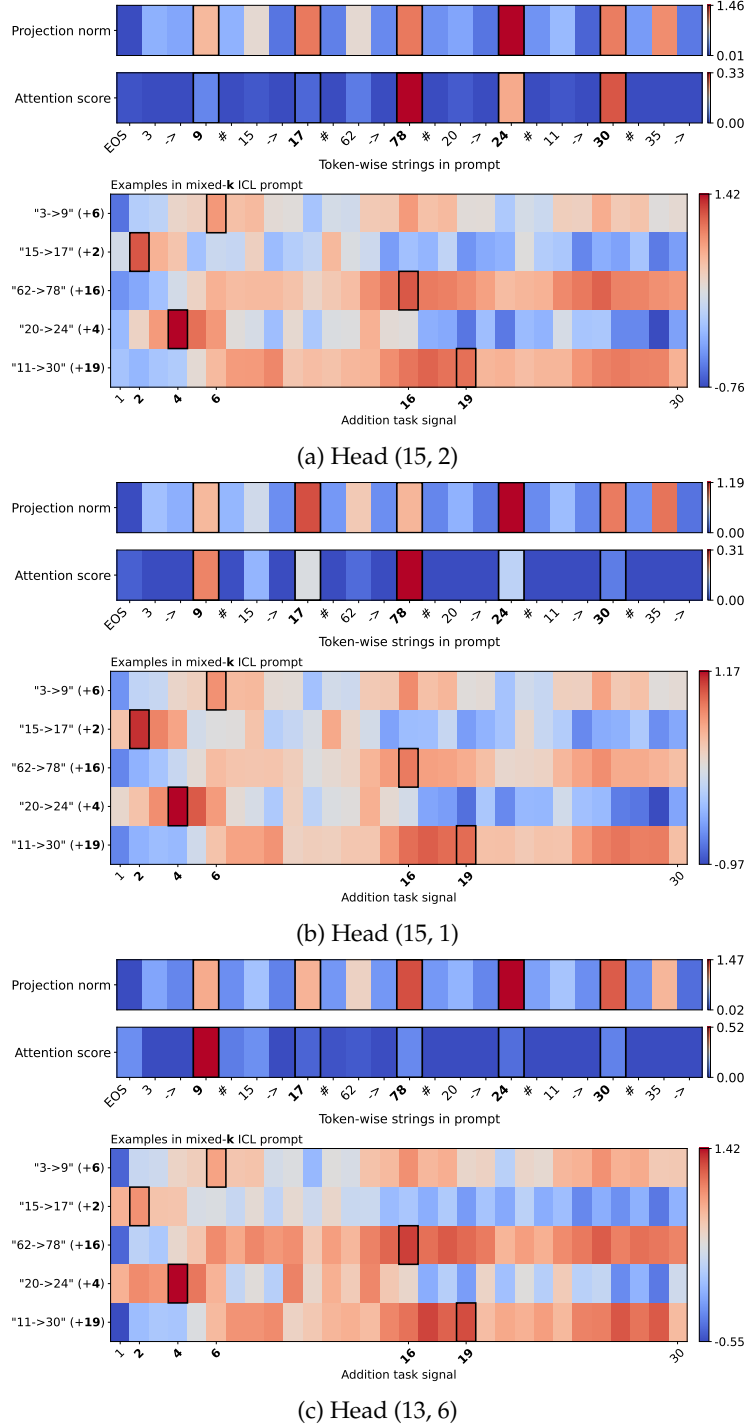


Figure 16: Llama-3-8B-instruct. For each head, the **top panel** shows the two factors of each previous token’s task-signal contribution $\alpha_t W_h O_h V_h z_t$: (1) the norm of the extracted information $\|W_h O_h V_h z_t\|$ and (2) the attention weight α_t ; both consistently peak at the tokens $t = y_i$ (in bold). The **bottom panel** shows the inner product between the projected signal from each y_i and the head vector $\tilde{h}_k$ (normalized to unit norm), which peaks at $k = y_i - x_i$, indicating that each head extracts $y_i - x_i$ from its corresponding demonstration $x_i \rightarrow y_i$.